

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

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MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

.....
COMMISSION NATIONALE D'ORGANISATION DE
L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM
(HND)
.....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

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MINISTRY OF HIGHER EDUCATION

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NATIONAL COMMISSION FOR THE
ORGANISATION OF HIGHER NATIONAL
DIPLOMA EXAM (HND) EXAM
.....

National Exam of Higher National Diploma 2024 Session

Speciality/Option: SOFTWARE ENGINEERING (SWE)

Paper : System Analysis and Design

Duration : 4 hours .

Credit : 7

ANSWER ALL QUESTIONS

SECTION A : OBJECT MODELING (25 MARKS)

A) MCQs 15 Marks)

One (01) mark for each correct answer

1. Which of the following is a structural model that demonstrates the other systems in the environment of the system being developed?
 - a) system context model
 - b) interaction model
 - c) environmental model
 - d) both system context and interaction
2. Which of the following come under system control?
 - a) Reconfigure
 - b) Shutdown
 - c) Powersave
 - d) All of the mentioned
3. Which model describes the static structure of the system using object classes and their relationships?
 - a) Sequence model
 - b) Subsystem model
 - c) Dynamic model
 - d) Structural model
4. Which model shows the flow of object interactions?
 - a) Sequence model
 - b) Subsystem model
 - c) Dynamic model
 - d) Both Sequence and Dynamic model

5. Which message is received so that the system moves to the Testing state, then the Transmitting state, before returning to the Running state?
 - a) signalStatus()
 - b) remoteControl()
 - c) reconfigure()
 - d) reportStatus()
6. A description of each function presented in the DFD is contained in a _____.
 - a) data flow
 - b) process specification
 - c) control specification
 - d) data store
7. Which diagram indicates the behaviour of the system as a consequence of external events?
 - a) data flow diagram
 - b) state transition diagram
 - c) control specification diagram
 - d) workflow diagram
8. A data model contains
 - a) data object
 - b) attributes
 - c) relationships
 - d) all of the mentioned
9. _____ defines the properties of a data object and take on one of the three different characteristics.
 - a) data object
 - b) attributes
 - c) relationships
 - d) data object and attributes
10. The _____ of a relationship is 0 if there is no explicit need for the relationship to occur or the relationship is optional.
 - a) modality
 - b) cardinality
 - c) entity
 - d) structured analysis
11. A _____ is a graphical representation that depicts information flow and the transforms that are applied as data moves from input to output.
 - a) data flow diagram
 - b) state transition diagram
 - c) control specification
 - d) workflow diagram
12. . A data condition occurs whenever a data is passed to an input element followed by a processing element and the result in control output.
 - a) True
 - b) False

13. The _____ enables the software engineer to develop models of the information domain and functional domain at the same time
- a) data flow diagram
 - b) state transition diagram
 - c) control specification
 - d) activity diagram
14. The _____ contains a state transition diagram that is a sequential specification of behavior.
- a) data flow diagram
 - b) state transition diagram
 - c) control specification
 - d) workflow diagram
15. Which of the following is not a construct?
- a) sequence
 - b) condition
 - c) repetition
 - d) selection

B) Structural (10 Marks)

1. How do you create use case diagrams? (3 marks)
2. Explain the following terms: actor, use case, system boundary, and relationship. Use layperson's language, as though you were describing them to a user. (3 marks)
3. Which of the following could be an actor found on a use case diagram? Why?
 - a) Ms. Mary Sara
 - b) Supplier
 - c) Customer
 - d) Internet customer
 - e) Mr. John Sabi
 - f) Data entry clerk
 - g) Database administrator
 (4 marks)

SECTION B: DATABASE ADMINISTRATION WITH MYSQL (25 MARKS)

1.
 - a) Name three ways null values can be interpreted in a relational database.
 - b) Why is this problematic?
 (3+4 = 7 marks)
2. What are the two dimensions in which to optimize a relational database? (3 marks)
3. What is the purpose of normalization? (2 marks)
4. How does a model meet the requirements of third normal form? (3 marks)

5. Describe three situations that can be good candidates for denormalization. (6 marks)
6. Describe several techniques that can improve performance of a database. (2 marks)
7. What is an index and how can it improve the performance of a system? (2 marks)

SECTION C: WEB PROGRAMMING (25MARKS)

1. What are the advantages of JSP over Servlet? (3 marks)
2. What are the added benefits of HTTP/2 compared to HTTP 1.1? (3 marks)
3. List three (03) ways to speed up Page Loading. (3 marks)
4. How is XHTML different from HTML? (3 marks)
5. List out newly introduced input types, APIs, form elements, and elements that support media content in HTML5. (5marks)
6. While building a web application, how do you consider SEO, maintainability, UX, performance, and security? (5 marks)
7. Describe concept of *Type Coercion* in JavaScript. (3 marks)

SECTION D: MOBILE PROGRAMMING (25MARKS)

1.
 - a) How can you store data in Android and iOS?
 - b) Explain the proper way of setting up an Android-powered device for application development. (2+3 = 5 marks)
2. State the different types of mobile terminals that you know. (2 marks)
3. How can you debug your app when it's already released? (2 marks)
4. Describe the android framework. (3 marks)
5. List the four key components of android architecture (2 marks)
6. What is an android operating system? (1 mark)
7.
 - a) Differentiate between native apps, hybrid apps and web apps.
 - b) What are the advantages of native application development? (2+2 = 4 marks)
8. What Is the Google Android SDK? (2 marks)
9. What is the importance of having an emulator within the Android environment? (4 marks)